

Hima — QA Release Testing SOP & Roadmap

Standard operating procedure for testers: what to test **before** release and **after** release
Innovfix Pvt. Ltd. · Hima App · v1 · 2026-07-14 · Owner: Yuvanesh (Tech Lead)

How to use this doc. For every release: run **Phase 1–4 (before release)** and only ship when the sign-off gate passes; then run **Phase 5 (after release)** on production. **BEFORE** = pre-release on staging/APK. **AFTER** = post-release on live prod. **MUST-PASS** = a failure blocks release.

1. Testing Phases (the process)

Phase	When	What
0 · Build sanity	New APK arrives	Installs, opens, no crash on launch, correct version number, login screen loads.
1 · Regression	Before release	Every module's core flows still work (Section 3 checklists).
2 · New-feature	Before release	Deep-test whatever changed in THIS release (from the branch/changelog).
3 · Critical-path	Before release	Money + calls deep test (Section 4) — zero tolerance.
4 · Sign-off gate	Before release	No Sev-1/Sev-2 open. Section 6 checklist signed.
5 · Post-release smoke + monitor	Right after release	Prod smoke on 2–3 real devices + watch live metrics for 2–4h (Section 5).
6 · Rollback watch	First 24h	If a rollback trigger hits (Section 5), revert/hotfix immediately.

Environments & test accounts (set up once)

- Devices:** ≥2 Android phones (1 low-end, 1 high-end), on WiFi **and** mobile data. Test the **Play Store / Internal Testing** build, not only sideload (sideload can pass while Play build fails — always confirm on the real distribution channel).
- Accounts:** a **male caller** (with test coins), a **female creator** (verified, with bank/UPI + PAN), and a **fresh new user** (for onboarding). Keep a second creator for block/DND pairing tests.
- Money:** use small real amounts (₹1–₹49) for real gateway tests — never assume; a coin credit must be verified end-to-end.

2. Bug severity & the release gate

Severity	Examples	Blocks release?
Sev-1 Critical	Coins credited without payment / free coins · double billing · double payout · app crash on launch · calls don't connect · can't log in · money lost	🔴 YES — hard block
Sev-2 Major	A gateway fails · withdrawal stuck · matcher connect-rate drops · KYC/verify broken · chat not delivering	🔴 YES
Sev-3 Minor	UI glitch, wrong text, non-blocking edge case	🟡 Ship with a follow-up ticket

3. BEFORE RELEASE — module-by-module roadmap **BEFORE**

3.1 Login · Auth · Onboarding

✓ Test
☐ New user: enter mobile → OTP received → verify → profile setup → lands on home.
☐ Wrong OTP rejected; resend OTP works; OTP expiry handled.
☐ Existing user login + logout + re-login (token refresh) works.
☐ Profile update: name (one-time name-change rule), avatar, interests (optional), language.

✓ Test

- Gender selection correct; male vs female home screens differ correctly.
- AI onboarding chat triggers for new male users; icebreaker shows.
- Force-update: a below-min-version user is prompted/blocked per `individual_app_update`.
- Delete account flow works; deleted user can't log back in.
- **NEW** Keyboard visibility: after entering the mobile number, the **mobile number field is fully visible** above the keyboard (not hidden); and on the OTP step the **6-digit OTP boxes are fully visible** above the keyboard. Both steps, low-end + high-end device. (QA #4)

3.2 Creator verification / KYC

✓ Test

- Voice verification: record voice → female auto-approve / male-voice reject.
- PAN KYC (Cashfree Secure-ID, PaySprint fallback): valid PAN passes, invalid rejected.
- Bank + UPI details save + validate; can't withdraw before verification.

3.3 Calls (audio + video) **MUST-PASS parts**

✓ Test

- **Random match:** caller taps random → matched to an online creator in his language → rings.
- **Connect:** creator answers → Agora audio/video connects both ways → timer starts (`started_time`).
- **Billing:** coins deduct per minute from caller; creator income accrues correctly; amounts match the rate.
- **No double-billing** (rapid reconnects / weak network don't charge twice).
- **Coins exhaust:** server force-ends the call when caller's coins hit 0.
- **Direct call** to a specific creator; recent/favourite call works.
- **Reject:** creator rejects → caller sees "busy/unavailable"; 3-reject cool-down applies.
- **Missed call:** unanswered → `missed_call` count + push to creator.
- **Video & audio both;** switch between; low-network behaviour (call doesn't hang forever).
- Call ends cleanly on either side; end reason recorded; earnings shown to creator.
- **NEW** **Network drop while connecting:** caller's network dies mid-ring → the **creator's ring auto-ends** within ~45s (no connected-but-silent call that has to be hung up manually). (QA #2)
- **NEW** **Ring stops on cancel — app closed:** creator's app is closed/backgrounded and rings → caller cancels → the ringing/vibration **stops immediately** (no ghost ring to dismiss by hand). (QA #6)
- **NEW** **Call history label:** a connected + completed call always shows its **duration** (even a very short call) in Call History — it is **never** labelled "Missed". Genuinely unanswered calls still show "Missed". (QA #10)
- **NEW** **No duplicate/ghost calls on reopen:** after the creator closes and reopens the app, previously received incoming/missed calls do **not** re-appear or re-ring as new incoming calls (queued pushes for already-ended calls are dropped). (QA #7)

3.4 Do-Not-Disturb & Blocking

✓ Test

- Creator turns DND on → she gets NO incoming calls/chat pings → shows unavailable → stays on until she turns it off.
- Call-block: blocked caller can't call her at all; matcher never pairs them again.
- Chat-block: blocked caller can't message; existing chat stops.
- Both honoured silently (no "you are blocked" popup) across call + chat + matcher.
- **NEW** After blocking a contact, that conversation **moves to the bottom** of the chat list (and shows the Blocked badge). (QA #8)

3.5 Coins & Payments MUST-PASS

Zero tolerance: coins must be credited ONLY after a verified real payment, exactly once. Test EACH live gateway with a small real amount.

✓ Test

- ☐ **Cashfree:** buy a pack → pay → coins credited exactly once; check `transactions.add_coins`.
- ☐ **PhonePe:** same end-to-end.
- ☐ **Razorpay / UPI link:** same end-to-end.
- ☐ **Payment FAIL / cancel:** NO coins credited; balance unchanged; clean error to user.
- ☐ **Webhook replay / forged:** a repeated or unverified webhook does NOT re-credit or free-credit coins.
- ☐ Offers, trial coins, coupons apply the correct discounted price + coins.
- ☐ Gateway switch/maintenance flag: when a gateway is off, the app falls back correctly.

3.6 Autopay / Subscription

✓ Test

- ☐ ₹1 trial sheet shows to never-subscribed; mandate created (UPI, GPay-compatible).
- ☐ Daily coin claim works; recurring ₹299 charge fires; subscription status correct.
- ☐ Cancel subscription works; CHARGE-fail vs AUTH-fail push messages correct.

3.7 Withdrawals / Payouts MUST-PASS

Zero tolerance: a creator must be paid the correct amount exactly once — never double-paid, never a phantom deduction.

✓ Test

- ☐ Request withdrawal → balance deducts → Cashfree payout (IMPS) → status Pending→Paid → money reaches bank.
- ☐ **No double-pay** (rapid taps / retries don't pay twice).
- ☐ Failed/rejected transfer → balance rolled back correctly.
- ☐ PAN gate + TDS applied where required; withdrawal blocked without PAN.
- ☐ Min-amount, bank-detail-missing, and reversal (auto-refund coins) edge cases.

3.8 Creator earnings, tiers & bonuses

✓ Test

- ☐ Income from calls shows correctly on creator dashboard; balance + `total_income` match.
- ☐ Tier/score updates; call-bonus tiers pay the right bonus; icebreaker prompts show.
- ☐ Creator warnings issue + auto-disable + auto-unblock work; star-creator rates apply.
- ☐ **NEW** **Call-duration match:** the creator's **Call History duration = Earnings/Transactions duration = the real connected talk time** (measured answer→hangup, with no ring/setup time included). (QA #12)
- ☐ **NEW** **Every connected call is credited:** a call that connected (`started_time` set) credits the creator for its real talk time **even if the caller's app didn't cleanly report the end** (force-close / network drop) — no answered call is left with ₹0 income. (QA #11)

3.9 Chat & Social (real-time)

✓ Test

- ☐ Send/receive message in real time (both directions); single→blue ticks (read receipts).
- ☐ Reactions add/update; message delete (archived server-side) + reflects for both.

✓ **Test**

- ☐ Bad-word message filtered/handled; presence/typing indicators show.
- ☐ Attachments upload; gifts send + deduct coins + credit creator; friend requests flow.
- ☐ Open-chat-from-notification deep-links to the right thread.
- ☐ **NEW** Image and audio **attachment options are removed** from the chat composer (emoji available in their place). (QA #3)

3.10 Notifications

✓ **Test**

- ☐ Push received (FCM + OneSignal): incoming call, new message, friend request, creator-online.
- ☐ Tapping a notification opens the correct screen; DND suppresses pings.
- ☐ Notification click/conversion tracking records (if part of this release).
- ☐ **NEW** Tapping a **video**-call notification starts a **video** call (not audio) when both call types are enabled. (QA #9)
- ☐ **NEW** Rejected calls are **not** shown/pushed as "missed"; missed-call notification text shows the correct call status (not both audio & video). (QA #5)

3.11 Growth, Games & misc

✓ **Test**

- ☐ Install referrer / tracking link / deep link attribute correctly (if changed).
- ☐ IPL rooms create/join/leave/react; Ludo invite + play (if in release).
- ☐ Reports/ratings/reviews; app-store rating prompt triggers per rules.

3.12 Admin panel (backend) **BEFORE**

✓ **Test**

- ☐ **NEW** Block a user/creator with a reason → reopen the account → the **block reason and blocked status/icon are still shown** (retained for audit). (QA #1)

4. CRITICAL MUST-PASS scenarios (money + calls) **MUST-PASS**

These four must be re-verified **every release**, end-to-end, on the real Play build. A single failure = **do not ship**.

1. **Pay → get coins, once.** Real payment on each live gateway → exact coins credited once. Fail/forged → no credit.
2. **Call → correct billing.** Connect → per-minute deduct + creator income correct → no double-bill → force-end on 0 coins.
3. **Withdraw → paid once.** Payout reaches bank, correct amount, no double-pay, rollback on failure.
4. **Matcher + safety.** Random call connects (~normal rate), and DND/blocked pairs are never matched.

5. AFTER RELEASE — production verification & monitoring **AFTER**

5.1 Smoke test on prod (first 30 min, 2–3 real devices)

✓ **Test on the LIVE app**

- ☐ Fresh install from Play → open → login/OTP works.
- ☐ One real random call connects both ways.
- ☐ One real small coin purchase credits coins.
- ☐ One real chat message delivers + blue ticks.

✓ **Test on the LIVE app**

- ☐ A test withdrawal request is accepted (status pending).

5.2 Watch these live metrics for 2–4 hours

Metric	Healthy	Rollback trigger
Connect rate (started_time)	~45–52%	Sudden drop >10 pts vs same slot yesterday
Revenue (add_coins/hr)	≈ prior day same hour	Sharp fall vs baseline (not a demand dip)
Gateway success %	Cashfree ~56%, PhonePe ~51%	Any gateway drops >10 pts
Payout %	~40%	Spikes (income leak / ghost calls)
Crashlytics crash-free	>99%	New crash cluster after release
Error log (laravel.log)	baseline noise	New error spiking (e.g. class-not-found, 500s)

Version-aware check: compare the NEW version's users (by `current_version`) vs the stable version on connect rate, ARPU, and gateway success — a bad release shows as the new version underperforming (like v1109 did). Roll out in staged % and watch before 100%.

5.3 Rollback / hotfix criteria

- Any **Sev-1** in production (money loss, free coins, double-pay, mass crash, calls not connecting) → **halt rollout + hotfix/rollback immediately**.
- Connect rate or revenue drops sharply and it's traced to the release (not external) → roll back the staged %.
- A gateway or payout breaks → flip its kill-switch (`gateway_config`) and fix.

6. Sign-off checklist (gate before release)

✓ **Item**

- ☐ Build sanity passed (installs, opens, correct version, no launch crash) — on the **Play/Internal** build.
- ☐ All 4 **MUST-PASS** critical scenarios passed end-to-end.
- ☐ Regression (Section 3) done for all modules; new-feature deep-tested.
- ☐ **Zero open Sev-1 / Sev-2**. All logged in the bug tracker with severity.
- ☐ Tested on ≥2 devices, WiFi + mobile data.
- ☐ Post-release smoke + monitoring plan assigned (who watches, for how long).
- ☐ Rollback owner + kill-switches identified. Signed by QA lead + Tech lead.

5. AFTER RELEASE — production verification & monitoring AFTER

5.1 Smoke test on prod (first 30 min, 2–3 real devices)

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Local Call-Audio Extraction Foundation

SOP revision 1.1 - 2026-07-17 - Applies to development APK before any transcription provider is connected.

Release gate

ACX-001, 002, 003, 004, 005, 007 and 008 are MUST-PASS on real devices. A compile/unit-test pass alone does not authorize production rollout. Attach logs, profiler screenshots and defect/retest evidence.

ACX-001 - Male local microphone extraction	MUST-PASS / P0
Preconditions / build / role	Development APK; two real devices; male caller and female creator; call connects normally.
Test data	Speak a fixed 65-second script on the male device; female remains quiet.
Steps	1. Start an audio call. 2. Speak before and across 30 s and 60 s boundaries. 3. End normally. 4. Capture filtered AudioExtractor logs.
Expected result	Call audio remains clear. Male device reports monotonic chunks with 30 s max and 1 s overlap after the first. Female local track does not contain male playback. No file/network/AI/warning appears.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____
ACX-002 - Female local microphone extraction and attribution	MUST-PASS / P0
Preconditions / build / role	Same two-device build; connected audio call.
Test data	Female speaks fixed script while male stays quiet; then both alternate turns.
Steps	1. Record female-only speech for 35 s. 2. Alternate speakers. 3. Compare each device's local metadata and audible call.
Expected result	Female device marks local speech. Male device marks only its own spoken turns, not remote playback. No speaker-attribution crossover or call-audio regression.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____
ACX-003 - Manual mute privacy boundary	MUST-PASS / P0
Preconditions / build / role	Connected call; metadata logs available.
Test data	Speak A while live, mute and speak PRIVATE, unmute and speak B.
Steps	1. Speak A. 2. Tap mute. 3. Speak for at least 10 s while muted. 4. Unmute and speak B. 5. End call.
Expected result	Peer hears A and B only. Pre-mute partial closes; no extracted duration advances during mute; post-unmute starts a clean non-overlapping boundary. PRIVATE is never present in any future payload.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____
ACX-004 - Phone/audio-focus interruption	MUST-PASS / P0
Preconditions / build / role	Connected call; second incoming cellular or VoIP/audio-focus event available.
Test data	Speak before, during and after the interruption.
Steps	1. Trigger interruption. 2. Speak while HIMA is held. 3. End interruption. 4. Resume HIMA speech. 5. Repeat once.
Expected result	Extraction pauses with outgoing mic, resumes with a clean boundary, sequence remains monotonic, call does not crash, duplicate workers are not created and held/private speech is excluded.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____

Long-Call, VAD and Failure Coverage

Environment: development APK, two real devices, Wi-Fi and mobile data, wired/Bluetooth where available.

ACX-005 - Boundary and hour-long stability	MUST-PASS / P0
Preconditions / build / role	Two charged test accounts; devices on power; Android Studio profiler or equivalent.
Test data	At least 65 min call; scripted words spanning each sampled 30 s boundary.
Steps	1. Run call for 65 min. 2. Sample logs at start, 30 min and 60 min. 3. Exercise speaker, wired and Bluetooth routes. 4. End normally.
Expected result	No 30 s recording stop, missing tail, duplicate-only final chunk, ANR, OOM or increasing worker count. Windows stay <=960 KB; sequence/timestamps are monotonic; audible call quality remains normal.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____
ACX-006 - VAD shadow labels across languages	P1
Preconditions / build / role	Connected call in a quiet room, then normal background noise.
Test data	English plus all 11 HIMA languages and code-switching; silence; soft speech; loud speech; fan noise.
Steps	1. Speak each sample naturally. 2. Include quiet and noisy periods. 3. Compare hasSpeech/voiced ratio to the known speaking periods.
Expected result	VAD is language-independent and labels ordinary speech without modifying/cutting PCM. Silence is normally false. Noise false positives are documented; any speech false negative blocks using VAD to skip future API calls.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____
ACX-007 - Teardown, reconnect and permission failure	MUST-PASS / P0
Preconditions / build / role	Connected call; ability to toggle network and revoke RECORD_AUDIO.
Test data	Wi-Fi/mobile switch, temporary loss, peer leave, local hangup, permission revoke.
Steps	1. Reconnect network twice. 2. End via peer and local paths. 3. Repeat with permission revoke. 4. Start a fresh call after each case.
Expected result	Observer is not duplicated on reconnect; dispose is safe from leaveChannel/onDestroy; no extractor thread survives teardown; next call captures normally; existing call failure handling remains authoritative.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____
ACX-008 - Privacy, no-spend and regression gate	MUST-PASS / P0
Preconditions / build / role	Fresh development APK; proxy/file inspection tools; admin access for read-only verification.
Test data	One 2-minute call containing harmless test speech.
Steps	1. Inspect app private cache/database before/during/after. 2. Inspect network requests. 3. Check admin/warnings. 4. Verify call settlement/rating flow.
Expected result	Zero audio files, uploads, transcript rows, provider calls, AI cost, admin audio, moderation results or warnings. Existing connection, mute, timer, hangup, settlement and rating flows still complete.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____

Creator-online Notification Call-Type Routing

SOP revision 1.2 - 2026-07-17 - Production payload hotfix. Applies to both instant and scheduled OneSignal creator-online senders.

Release gate	CTN-001, CTN-002 and CTN-003 are MUST-PASS. Do not send a production test notification without owner approval. Record payload evidence, device evidence, defect ID and retest status.
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CTN-001 - Instant video notification routes to video		MUST-PASS / P0
Preconditions / build / role	Backend hotfix on all three nodes; instant flag ON; current production APK; male caller and verified female creator; creator has both audio and video enabled.	
Test data	Use an approved test creator/caller pair. Enable video last so the instant alert advertises video.	
Numbered steps	1. Capture the generated OneSignal data object without exposing tokens. 2. Confirm call_type=video. 3. Tap the alert on the male device. 4. Observe the connecting and in-call screen.	
Expected result	Payload contains source=creator_online, the intended creator ID and call_type=video. Tap opens the video connecting flow and joins a video call; it never starts audio.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

CTN-002 - Scheduled fallback preserves video type		MUST-PASS / P0
Preconditions / build / role	QA/demo environment or an explicitly approved controlled production test; instant path disabled or deliberately made to leave the queue; scheduled creator processor active; both creator call types enabled.	
Test data	Queue one creator-online video event for the approved pair.	
Numbered steps	1. Let creators:process-notifications claim the queue row. 2. Capture redacted payload evidence. 3. Confirm call_type=video. 4. Tap the resulting alert.	
Expected result	Scheduled payload and human text agree on video. Tap opens video. The fallback no longer loses call_type or silently selects audio.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

CTN-003 - Audio notification remains audio		MUST-PASS / P0
Preconditions / build / role	Current APK; creator has both types enabled; approved instant and scheduled test paths available.	
Test data	Enable audio last so the advertised type is audio.	
Numbered steps	1. Exercise instant sender and scheduled sender separately. 2. Verify call_type=audio in each redacted payload. 3. Tap each notification.	
Expected result	Both alerts open the audio connecting flow. Video routing fix does not invert or break existing audio behavior.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

CTN-004 - Availability and stale-type fallback		P1
Preconditions / build / role	Current APK; controlled accounts; ability to change creator audio/video availability before tap.	
Test data	Video-only, audio-only, both-off, and a payload whose advertised type becomes unavailable before tap.	
Numbered steps	1. Test video-only and audio-only alerts. 2. Turn the advertised type off before tapping while leaving the other type on. 3. Test both-off. 4. Record resulting screen/message.	
Expected result	Available advertised type is honored. If it became unavailable, app uses the remaining available type. Both-off does not launch a billable call and shows the existing unavailable handling.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

CTN-005 - Legacy payload compatibility		P1
Preconditions / build / role	Debug/QA injection only; current APK; do not send a real provider notification.	
Test data	Creator-online payload omitting call_type, representing an old queued/provider payload.	
Numbered steps	1. Inject/open the legacy payload in QA. 2. Test audio-only, video-only and both-enabled creator states. 3. Confirm app remains stable and clears stored notification routing keys after use.	
Expected result	No crash. Audio-only and video-only still open their available type; both-enabled retains the documented backward-compatible audio fallback only for legacy payloads.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

CTN-006 - Production smoke and regression evidence		P1 / AFTER
Preconditions / build / role	Approved production smoke window; all three live hashes match; PHP-FPM 8.2/8.4 active; owner approval for any outward test notification.	
Test data	One approved audio alert and one approved video alert; no customer accounts or PII in evidence.	
Numbered steps	1. Verify live source/hash and service health. 2. Send only the approved test alerts. 3. Tap each on a current APK. 4. Monitor Laravel/PHP errors and creator-online send failure rate for 30 minutes. 5. Record rollback decision.	
Expected result	Audio opens audio and video opens video. No new PHP errors, send failures, duplicate sends, latency regression, database work or availability regression. Reverse the two payload lines if a rollback trigger occurs.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

Incoming Ring Stops After Caller Disconnect

SOP revision 1.3 - 2026-07-17 - Authenticated Redis heartbeat plus legacy fallback. No user_calls migration and no call-row write from liveness polling.

Release gate	RNG-001 through RNG-005 are MUST-PASS before rollout. Run network, Redis-failure and load cases in QA. Production calls or notifications require separate owner approval. Record build, timing, redacted evidence, defect and retest status.
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RNG-001 - Original ghost-ring reproduction on legacy APK MUST-PASS / P0	
Preconditions / build / role	QA backend patch; production-equivalent APK that does not send call_ring_heartbeat; caller and recipient test accounts; incoming screen polling active.
Test data	Audio and video, both call directions, caller network disabled during pre-answer setup.
Numbered steps	1. Start the call. 2. Confirm recipient is ringing. 3. Disable caller network before Answer. 4. Leave recipient untouched. 5. Measure from call creation and from network loss. 6. Repeat with caller app killed.
Expected result	Recipient ringtone, Telecom call and incoming notification all stop by 47.5 seconds from creation. No Answer screen remains and no old ring reappears. No coins are deducted because started_time remains empty.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____

RNG-002 - Authenticated heartbeat stops a disconnected caller quickly MUST-PASS / P0	
Preconditions / build / role	QA backend patch; new APK containing authenticated heartbeat; valid JWT; clocks synchronized for evidence only.
Test data	Drop Wi-Fi, mobile data, airplane mode and process kill separately at 5-10 seconds of ringing.
Numbered steps	1. Start ringing and capture successful heartbeat responses. 2. Trigger one disconnect mode. 3. Measure last accepted heartbeat to recipient teardown. 4. Repeat each mode for audio/video and both caller genders.
Expected result	Recipient teardown occurs in 15-18 seconds after the last accepted heartbeat. check_call_alive returns alive=false, ringable=false and reason=not_answered. The call row is not changed by the liveness poll.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____

RNG-003 - Answer and heartbeat timeout boundary MUST-PASS / P0	
Preconditions / build / role	QA backend patch and new APK; controllable latency or packet loss; both devices online initially.
Test data	Answer attempts immediately before and after the 15-second stale boundary, including a delayed heartbeat response.
Numbered steps	1. Delay caller heartbeat traffic. 2. Tap Answer at 14, 15 and 16 seconds in separate calls. 3. Observe accept-in-flight behavior, Agora join, server response and billing state. 4. Repeat 20 times.
Expected result	An Answer already in flight is not dismissed by a late poll callback. A call declared unreachable does not open a billable empty room. No row contains contradictory started and terminal state; no double ring occurs.
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____

RNG-004 - Heartbeat authorization and ownership		MUST-PASS / P0
Preconditions / build / role	API test client in QA; one male-initiated and one female-initiated call; caller, recipient and unrelated JWTs.	
Test data	Missing token, invalid token, recipient token, unrelated token, caller token, invalid and nonexistent call IDs.	
Numbered steps	1. POST each credential/call combination. 2. Verify status and response. 3. Inspect Redis keys without exposing tokens. 4. Confirm no user_calls write occurred.	
Expected result	Only the actual initiator can refresh the heartbeat. Missing/invalid authentication is rejected, recipient/unrelated users receive forbidden, malformed IDs are rejected, and no database row is mutated.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

RNG-005 - Delayed push, recovery and no resurrection		MUST-PASS / P0
Preconditions / build / role	QA backend patch; new APK; ability to delay FCM delivery and background/foreground the recipient app.	
Test data	Fresh ring, heartbeat-stale ring, 30-second age-guard ring and 45-second legacy-terminal ring.	
Numbered steps	1. Delay each incoming push. 2. Trigger foreground pending-call recovery. 3. Deliver the old push after each boundary. 4. Reopen the app. 5. Attempt a new unrelated call.	
Expected result	Only a fresh unanswered call is surfaced. ringable=false blocks stale/started calls, pending recovery does not resurrect them, and a later new call rings normally.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

RNG-006 - Redis failure and backward-compatible fallback		P1
Preconditions / build / role	QA only; controlled Redis unavailability or injected Cache exception; legacy and new APKs.	
Test data	Cache read failure, write failure, recovery during ring and heartbeat endpoint timeout.	
Numbered steps	1. Inject each failure. 2. Start a call and attempt heartbeats. 3. Observe API status, logs and recipient ring. 4. Restore Redis and repeat.	
Expected result	Cache failure does not end a legitimate call early and does not return a server 500. The new client degrades to the 45-second legacy terminal response; recovery needs no database repair.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

RNG-007 - Capacity and database regression gate		P1 / LOAD
Preconditions / build / role	Staging or isolated production-equivalent environment; representative concurrency; query and Redis telemetry.	
Test data	At least 100 concurrent ringing sessions for five minutes plus current check_call_alive traffic.	
Numbered steps	1. Generate 2.5-second authenticated heartbeats and both-side liveness polls. 2. Record p50/p95/p99 latency, Redis ops, PHP CPU/RSS and DB queries. 3. Verify query plan. 4. Compare baseline.	
Expected result	Heartbeat is bounded to about 24 requests/minute per ringing caller, uses one caller-auth lookup plus one user_calls PK lookup and one Redis SET, creates no migration or call-row write, and does not materially regress API/DB capacity.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

RNG-008 - Three-node production smoke and rollback		P1 / AFTER
Preconditions / build / role	Separate owner approval for node patch, PHP-FPM reload and any real-device call; all three node hashes verified first.	
Test data	One approved legacy-client call and one approved new-client call; no customer data in evidence.	
Numbered steps	1. Verify route/service/controller hashes on app-1/2/3. 2. Verify PHP-FPM 8.2/8.4. 3. Run approved calls. 4. Monitor 4xx/5xx, Redis latency and PHP errors for 30 minutes. 5. Record rollback decision.	
Expected result	All nodes behave identically, ordinary calls connect normally, legacy teardown meets 47.5 seconds, new teardown meets 15-18 seconds, and no billing, availability, database-write or service-health regression appears.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

Call Moderation Compact Controls

Demo admin UI revision - 2026-07-17 - Blade-only presentation change; control routes and behavior remain unchanged.

CMR-UI-001 - Desktop one-line control rail		MUST-PASS / P0
Preconditions / build / role	Demo admin account with Call Moderation access; desktop viewport at least 1200px; moderation settings visible.	
Test data	Master On and Off states, auto-warning On and Off states, minimum version 0 and a positive version code.	
Numbered steps	1. Open Call Moderation Review. 2. Inspect the first control area at 1200px and 1920px widths. 3. Verify labels, values and state colors. 4. Expand Recent changes.	
Expected result	Moderation, Auto warning and Minimum app version appear in one compact row. Each control has a short label, current value and Save button. No former large headings or explanatory paragraphs remain. Recent changes stays available below the row without clipping or overlap.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

CMR-UI-002 - Control behavior and authorization regression		MUST-PASS / P0
Preconditions / build / role	Authorized demo admin; separate unauthorized or insufficient-role session; known initial control values; no real call required.	
Test data	Toggle each switch once and restore it; version values 0, valid positive integer, negative value and value above 2147483647.	
Numbered steps	1. Change each control and press its Save button. 2. Confirm the warning prompt. 3. Reload and verify persistence/history. 4. Test invalid version boundaries. 5. Repeat update requests without authorization.	
Expected result	Existing PATCH routes, CSRF, confirmations, permission checks, validation and audit history still work. Invalid values are rejected. Unauthorized users cannot update controls. Master, auto-warning and version semantics are unchanged, and all test values are restored after execution.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	

CMR-UI-003 - Responsive, keyboard and error-state QA		P1
Preconditions / build / role	Desktop and mobile browser/device; keyboard-only input; browser zoom 100% and 200%; ability to simulate validation error and infrastructure-disabled state.	
Test data	Viewports 390px, 768px, 900px and 1200px; long translated/admin font rendering where available.	
Numbered steps	1. Resize through each viewport. 2. Tab through switches, input, Save and Recent changes. 3. Test at 200% zoom. 4. Trigger a validation error. 5. Inspect disabled controls.	
Expected result	The controls remain one line on desktop and stack cleanly below 900px. Focus is visible, labels are announced, touch targets remain usable, errors remain visible, disabled controls cannot submit, and the evidence locator and call list retain their prior layout and behavior.	
Execution	Status: NOT RUN Evidence: _____ Defect: _____ Retest: _____	